

4E1301

Roll No. _____

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B.Tech. IV Sem. (Main/Back) Examination, July - 2023
Artificial Intelligence and Data Science
4AID2-01 Discrete Mathematics Structure
CS, IT, AID, CAI

Time : 3 Hours**Maximum Marks : 70****Instructions to Candidates:**

Attempt all ten questions from Part A. Attempt any five questions out of Seven questions from Part B and three questions out of Five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (As Mentioned in form No. 205).

PART - A**(Word limit 25)****All questions are compulsory.****(10×2=20)**

1. Prove that $A - B = B' \cap A'$. (2)
2. Define partial ordering relation with an example. (2)
3. Draw the truth table for $p \wedge q \rightarrow p \vee q$. (2)
4. Explain conjunctive normal form. (2)
5. Draw the Hasse diagram of the poset $(A, \leq)$ where $A = \{1, 2, 3, 4, 12\}$ and the partial order of divisibility on A is $a \leq b$ (i.e. if a divides b). (2)
6. 8 boys and 5 girls constitute a group. In how many ways seven of them can be selected if the selections always have atleast 3 boys and 2 girls. (2)
7. If a, b are any elements of a group G, then prove that $(ab)^{-1} = b^{-1}a^{-1}$. (2)
8. If a is a generator of a cyclic group, then prove that a^{-1} is also its generator. (2)
9. Draw graph which is
 - a. Eulerian but not Hamiltonian.
 - b. Hamiltonian but not Eulerian.
10. Define chromatic number. (2)

Part - B

(Word limit 100)

Attempt any five questions

1. Derive an expression for total power of an AM wave in terms of carrier power and modulation index. (5×4=20)
2. Explain the operation of TDM with the help of suitable diagram and application.
3. Discuss the DPCM technique with the help of suitable diagram.
4. Explain the terms 'Line coding' and 'Inter - symbol interference' in a digital modulation.
5. Compare the FHSS with DSSS using suitable diagrams.
6. Draw the block diagram of a PLL and explain its working principle in brief?
7. Explain the terms 'Aliasing' and 'Interpolation' in Pulse Analog Modulation.

Part - C

(Descriptive/Analytical/Problem solving/Design questions)

Attempt any Three questions

(3×10=30)

1. An antenna has an impedance of 40 Ω . An unmodulated AM signal produces a current of 4.8 A. The modulation is 90 percent. Calculate (a) the carrier power, (b) the total power, and (c) the sideband power. The transmitter experiences an antenna current change from 4.8 A unmodulated to 5.1 A. What is the percentage of modulation?
 2. Compare PAM, PWM and PPM modulation schemes with the help of suitable applications.
 3. Derive the expressions for signal to quantization noise ratio in PCM and Delta modulation techniques.
 4. Explain the following digital modulation techniques (any two);
 - i. MSK.
 - ii. FSK.
 - iii. QPSK.
 5. Write short notes on the following (any two):
 - i. CDMA.
 - ii. Coherent BPSK.
 - iii. Pseudo - Noise Sequences.
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